Calc AB: Integrating Powers of Sine and Cosine

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DIRECTIONS *Evaluate each integral or simplify the expression. For ODD powers: peel one factor and use Pythagorean substitution. For EVEN powers: apply the half-angle identities in the reference box below. Problems 12-15 review integration, limits, and derivatives.*

ODD POWERS | Peel one factor, then substitute:	**EVEN POWERS | Use half-angle identities:**
$\sin^2 x = 1 - \cos^2 x$ $\cos^2 x = 1 - \sin^2 x$	$\sin^2\theta = \frac{1}{2}(1 - \cos 2\theta)$ $\cos^2\theta = \frac{1}{2}(1 + \cos 2\theta)$

1 Evaluate:

$$\int \cos^2(2x)\,dx$$

Answer: ____________________

2 Evaluate:

$$\int \sin^3(3x)\,dx$$

Answer: ____________________

3 Evaluate:

$$\int \cos^4 x\,dx$$

Answer: ____________________

4 Evaluate:

$$\int \sin^2\left(\frac{x}{2}\right)dx$$

Answer: ____________________

5 Evaluate:

$$\int \sec^2(4x)\,dx$$

Answer: ____________________

6 Evaluate:

$$\int \sin^2 x \cos x\,dx$$

Answer: ____________________

7 Evaluate:

$$\int 5\sin^2(2x)\,dx$$

Answer: ____________________

8 Evaluate:

$$\int \cos^3 x \sin x\,dx$$

Answer: ____________________

9 Evaluate:

$$\int_0^{\pi/2} \cos^3 x\,dx$$

Answer: ____________________

10 Evaluate:

$$\int_0^{\pi/2} \sin^2 x\,dx$$

Answer: ____________________

11 Evaluate:

$$\int \tan^2(4x)\,dx$$

Answer: ____________________

12 Given:

$f'(x) = 3x^2 - 2,\ f(2) = 14.$ Find $f(1)$.

Answer: ____________________

13 Evaluate:

$$\lim_{h\to 0}\frac{\sin(x+h) - \sin(x)}{h}$$

Answer: ____________________

14 Find f'(x) if:

$$f(x) = \ln\left(\frac{e^{\sin 3x}}{\cos x}\right)$$

Answer: ____________________

15 Find dy/dx if:

$$y = \tan^{-1}(3x)$$

Answer: ____________________

Answer Key & Solutions

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TEACHER NOTES *For integrals 1-11, verify answers by differentiating. Problem 9 ans=2/3; Problem 10 ans=pi/4. Problem 12: integrate f'(x) to get f(x), use f(2)=14 to find C=2, then f(1)=9. Problems 13-15 test limit definition, log differentiation, and arctan derivative.*

1 Evaluate:

$\int \cos^2(2x)\,dx$

= $\frac{1}{2}x + \frac{1}{8}\sin(4x) + C$

Even: cos^2(2x) = (1+cos 4x)/2; integrate term by term.

2 Evaluate:

$\int \sin^3(3x)\,dx$

= $-\frac{1}{3}\cos(3x) + \frac{1}{9}\cos^3(3x) + C$

Odd: peel sin(3x), write sin^2 = 1-cos^2; u-sub u = cos(3x).

3 Evaluate:

$\int \cos^4 x\,dx$

= $\frac{3}{8}x + \frac{\sin(2x)}{4} + \frac{\sin(4x)}{32} + C$

Even: cos^4x = [(1+cos 2x)/2]^2; expand, then apply half-angle again.

4 Evaluate:

$\int \sin^2\left(\frac{x}{2}\right)dx$

= $\frac{1}{2}x - \frac{1}{2}\sin(x) + C$

Even: sin^2(x/2) = (1-cos x)/2; integrate directly.

5 Evaluate:

$\int \sec^2(4x)\,dx$

= $\frac{1}{4}\tan(4x) + C$

Direct antiderivative of sec^2; u = 4x gives a factor of 1/4.

6 Evaluate:

$\int \sin^2 x\cos x\,dx$

= $\frac{1}{3}\sin^3 x + C$

u-sub u = sin x, du = cos x dx. Gives int u^2 du = u^3/3.

7 Evaluate:

$\int 5\sin^2(2x)\,dx$

= $\frac{5}{2}x - \frac{5}{8}\sin(4x) + C$

Even: sin^2(2x) = (1-cos 4x)/2. Multiply by 5 and integrate.

8 Evaluate:

$\int \cos^3 x\sin x\,dx$

= $-\frac{1}{4}\cos^4 x + C$

u-sub u = cos x, du = -sin x dx. Gives -int u^3 du = -u^4/4.

9 Evaluate:

$\int_0^{\pi/2} \cos^3 x\,dx$

= $\frac{2}{3}$

Odd: peel cos x, write cos^2 = 1-sin^2; u = sin x; eval [u - u^3/3] from 0 to 1.

10 Evaluate:

$\int_0^{\pi/2} \sin^2 x\,dx$

= $\frac{\pi}{4}$

Even: sin^2x = (1-cos 2x)/2; eval [x/2 - sin(2x)/4] from 0 to pi/2.

11 Evaluate:

$\int \tan^2(4x)\,dx$

= $\frac{1}{4}\tan(4x) - x + C$

Identity: tan^2(4x) = sec^2(4x) - 1; integrate each term separately.

12 Given:

$f'(x) = 3x^2 - 2,\ f(2) = 14.$ Find $f(1)$.

= $f(1) = 9$

Integrate: f(x) = x^3 - 2x + C. Use f(2)=14 to find C=2. Then f(1) = 9.

13 Evaluate:

$\lim_{h\to 0}\frac{\sin(x+h) - \sin(x)}{h}$

= $\cos(x)$

This is the limit definition of the derivative: d/dx[sin x] = cos x.

14 Find f'(x) if:

$f(x) = \ln\left(\frac{e^{\sin 3x}}{\cos x}\right)$

= $3\cos(3x) + \tan(x)$

Simplify first: f(x) = sin(3x) - ln(cos x). Then differentiate each term.

15 Find dy/dx if:

$y = \tan^{-1}(3x)$

= $\frac{3}{1+9x^2}$

d/dx[arctan(u)] = u'/(1+u^2). Here u = 3x, u' = 3.